ROS 2 Turtlesim Follower Project

This repository contains the deliverables for **Lab Session 4** of the Mobile Robotics course. The project demonstrates the use of ROS 2 Launch files, Rosbag for data recording, and a custom Python node to implement a "Follow-the-Leader" behavior between two turtles in the Turtlesim simulator.

Project Structure

- `launch/turtlesim_launch.py` : Automates the startup of the simulator, teleop control, and spawning of the second turtle.
- `my_launch_pkg/follower_node.py` : The control logic that makes Turtle 2 follow Turtle 1.
- `README.md` : Project documentation.

Installation & Setup

1. **Clone the repository** into your ROS 2 workspace:

```
cd ~/ros2_ws/src
# git clone <your-github-link-here>
```

2. **Build the workspace**:

```
cd ~/ros2_ws
colcon build --packages-select my_launch_pkg
source install/setup.bash
```

How to Run

1. **Launch the Simulation**: Starts the environment and spawns the turtles.

```
ros2 launch my_launch_pkg turtlesim_launch.py
```

2. **Run the Follower Node**: In a new terminal, start the chasing logic:

```
source install/setup.bash
ros2 run my_launch_pkg follower_node
```

3. **Control the Leader**: Use the arrow keys in the teleop terminal to move Turtle 1. Turtle 2 will automatically follow.

Data Recording & Visualization

- **Recording**: To record the trajectory, run `ros2 bag record /turtle1/cmd_vel /turtle2/cmd_vel`.
- **Plotting**: Use `rqt_plot` to visualize `/turtle1/cmd_vel/linear/x`.

Control Logic

The follower uses a **Proportional (P) Controller**:

- **Linear Velocity**: $v = 1.5 \times \text{distance}$
- **Angular Velocity**: $\omega = 4.0 \times \text{angle\_error}$
- **Safety Distance**: The follower stops if it is within 1.0 unit of the leader.